

Automatic Calibration of DAS and Camera

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I. ABSTRACT

Distributed acoustic sensing (DAS) is a powerful tool thanks to its ease of use, high spatial and temporal resolution and sensitivity. Growing demand for long-distance DAS measurements, in conjunction with the development of efficient algorithms for data processing, has led to an increased interest in the technology from industry and academia. Machine-learning-based data processing, however, necessitates tedious calibration experiments that require significant effort and resources. Any form of processing of real DAS data requires time and space calibration with the objects that move above the surface nearby the fiber. If the fiber location documentation is missing or obsolete, the latter turns out to be a significant challenge that not yet had been addressed with an automatic solution. In this paper we will discuss and exhibit a novel geophysics-driven approach, enhanced by computer vision, signal processing and gradient based optimization to offer a solution of the spatial calibration of a fiber with its environment, specifically aiming to determine the fiber sampling locations with an accuracy of no more than 5 meters from the actual fiber locations while ensuring that the solution remains physically feasible.

II. INTRODUCTION

A. Fiber Optic Seismology

Fiber optic seismology is a technique that utilizes fiber optic cables to record the seismic wave field. It involves the measurement of minuscule phase changes in the backscattered light traveling through the cable and converting it to directional strain along the fiber. This innovative method offers numerous advantages, such as dense spatial-temporal measurements, utilizing existing fiber infrastructure, and the ability to cover large distances

with a single cable. Common applications of fiber optic seismology include earthquake detection and early warning systems, monitoring of underground reservoirs [14] and fracking operations, studying volcanic activity, and assessing structural integrity in civil engineering projects. Another application that is subject to active research is monitoring traffic in crowded areas through measuring and analyzing the signals from the fiber optic cables that are commonly deployed for communication purposes. One significant technology that enables the sensitive and accurate measurement of the fiber-optic cables is Distributed Acoustic Sensing (DAS). By generating pulses at one terminal of the fiber and analyzing the back-scattered light from various points along the fiber, DAS technology is capable of converting standard fiber-optic cables into arrays of thousands of individual sensors [6], which are sensitive to the seismic wave-field generated by anthropogenic and natural activity around them [3].

B. Theoretical Background

Seismic waves are elastic waves that travel through the Earth. They can result from earthquakes, landslides, explosions, surface urban activity, and more. The velocity of seismic waves depends on the elastic properties of the subsurface, its density, and type of seismic wave. Seismic waves can be categorized as surface waves, which travel along the Earth's surface, or body waves, which travel within the Earth's interior. Surface waves can be caused by various sources in urban areas. They are interface waves that travel along the free surface, but due to a skin-depth effect are influenced by the elastic properties of the subsurface. They are generally larger in amplitude and slower than body waves. They decay more slowly with distance compared to body waves [1], as they spread

cylindrically rather than spherically. The two main types of surface waves are Rayleigh waves and Love waves. As fiber-optic cables are commonly installed underground alongside traffic roads, the weight of passing vehicles causes the subsurface to deform [13]. This deformation affects the fiber-optic cable, resulting in a tiny yet measurable strain. As the car moves away from a certain point on the road, the changes in the strain field are recorded as a strain. The seismic waves that propagate as a result of vehicular activity carry energy mainly in quasi-static frequencies which are approximately $< 1[Hz]$, and therefore the strain changes mainly correspond to this spectral band. Previous works [13], [10], [4] show that the Flamant-Boussinesq approximation describes the quasi-static deformation of the subsurface caused by a theoretical point load at a given location. According to this approximation, the displacement u_x of a point x, y, z on the ground in x direction, as a response to a point load located at the origin is:

$$u_x(x, y, z) = \frac{F}{4\pi G} \frac{x}{r^2} \left(\frac{z}{r} + \frac{2\nu - 1}{1 + \frac{z}{r}} \right) \quad (1)$$

The symbols in this equation are:

- x, y, z are spatial coordinates in 3-D space in which the displacement is measured. Here we assume that the x axis is aligned with the deployment direction the fiber-optic cable, y axis is such that together with x they span the plane of the ground's surface, and z is the perpendicular depth beneath the surface.

- $r = \sqrt{x^2 + y^2 + z^2}$

- F is the force applied by the point load on the ground.

- G is the shear modulus of the ground, which reflects the sensitivity of the medium to be deformed as a result of disturbance, and is assumed to be spatially uniform.

- ν is Poisson's Ratio of the medium.

C. DAS Measurements

In practical terms, the DAS system measures the strain (or strain-rate) along the fiber rather than directly measuring the displacement u_x . Additionally, the measurements are not taken at individual points

along the cable, but represent an average over a certain distance known as the gauge length [13], denoted as L . Consequently, the recorded response captured by the DAS system should be modeled as a spatial derivative of the average displacement along the gauge length of eq.1:

$$\varepsilon_{\text{DAS}} = \frac{1}{L} \left(u_x \left(x + \frac{L}{2}, y, z \right) - u_x \left(x - \frac{L}{2}, y, z \right) \right) \quad (2)$$

Furthermore, since the DAS system that was used in our work was optically configured to measure the strain rate, the response of the medium should actually be differentiated with respect to time as well. In conclusion, the response of the fiber strain rate to a point load at the origin, as measured by DAS system, is modeled as:

$$\dot{\varepsilon}_{\text{DAS}} = \frac{1}{L} \left(\dot{u}_x \left(x + \frac{L}{2}, y, z \right) - \dot{u}_x \left(x - \frac{L}{2}, y, z \right) \right) \quad (3)$$

The DAS measurement procedure involves setting specific temporal and spatial resolutions for sampling the fiber. This configuration results in a 2-D array output, where one axis represents time and the other axis represents distance along the fiber. Each spatial sampling point along the fiber is referred to as a channel.

D. Related Work

Previous studies have primarily focused on investigating different aspects of traffic signal analysis using distributed acoustic sensing (DAS). Some researchers have explored roadside seismic interferometry as a mean to understand traffic patterns, specifically traffic flow [7], [8]. Additionally, image processing techniques have been employed to address the spatial and temporal correlations among signals collected from each sensor within the distributed sensing array [11]. Dealing with the challenges posed by DAS data has also been a subject of research, with deep learning emerging as a potential solution. Previous works, such as [12], [17], have suggested utilizing the inherent spatial and temporal coherence of DAS data to mitigate incoherent noise, thereby enhancing data quality for subsequent tasks. Another approach proposed the use of Generative Adversarial Networks to synthesize highquality DAS data [5], compensating for the scarcity of reliable data for neural network training.

In the context of vehicle detection and classification, a study employed acoustic frequency-based features and a Support Vector Machine (SVM) to classify individual vehicles into different groups [9]. Furthermore, deep learning techniques have been applied to extract valuable information from DAS data for traffic analysis purposes. For instance, [13], [16] attempted an unsupervised learning approach using the UNet architecture to interpret seismic data based on seismic model equations. Another work, outlined in [15], focused on estimating vehicle occupancy from DAS measurements to enable intelligent transportation applications.

III. METHOD

In our research, we leverage a DAS interrogator unit to capture seismic signals within its sensitivity range. The output of the interrogator unit consists of seismic signals recorded over a specific time period, presented as 2-D arrays. These arrays exhibit spatial and temporal correlations between neighboring samples, allowing them to be interpreted as gray-scale images - depicting us the real form of seismic interference in the fiber. Thus these arrays will serve as the ground truth labels of our project. On the other hand, utilization of computer vision techniques of depth estimation and object detection, combined with prior research and formulation of the strain caused by moving vehicles nearby the fiber - enables us to generate synthetic image of the DAS data. Obtaining similarity between the two images is essentially equal to unveiling the fiber's trajectory as the strain formulas presented above embody the location of the fiber relatively to objects nearby. Our data consists of street scene video recordings that were conducted concurrently with DAS recording of a fiber that is located nearby the road.

In a simplified manner, our system can be described by the diagram in Figure 1.

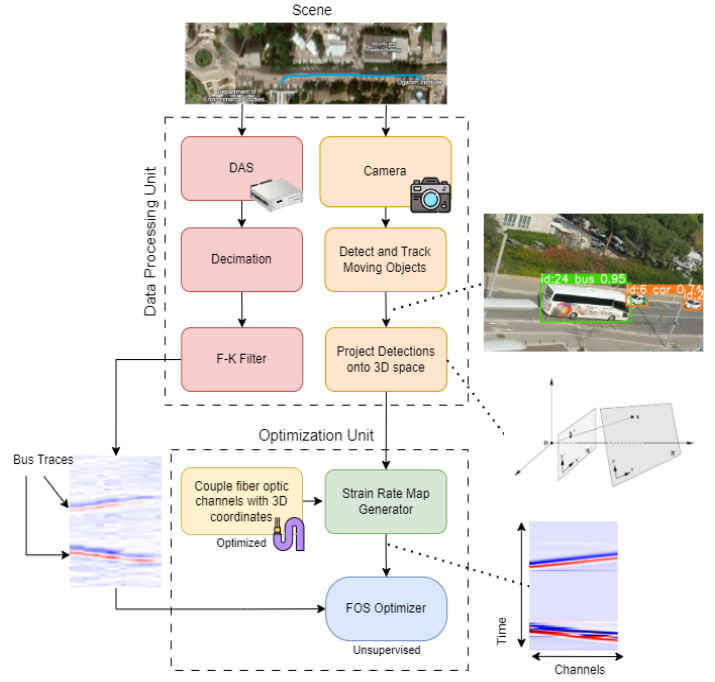


Fig. 1. Flattened System Design

- 1) **Optic-Fiber Signal Filtering:** The raw data obtained from the optic-fiber contains noise and irrelevant information. To prepare the input samples for the optimization process, decimation and F-K filtering are applied to enhance data quality and remove unnecessary information underlies in the recorded signal.
- 2) **Vehicle Tracking and Classification:** A video camera is utilized alongside the DAS interrogator to record the scene simultaneously. To extract all objects from the recorded video frames, the widely recognized YOLOv8 model is applied for tracking and classification. However, in order to generate strain rate maps that align with the optic-fiber space required additional actions to be taken:
 - **Removal of still objects:** YOLO detects all objects in the frame, while the DAS is sensitive to moving objects exclusively. Thus, standing objects must be removed to ensure consistency between the two

datasets.

- Spatial Interpolation of missing detections: YOLO often missdetects objects. Since our method is highly dependant on the differentiability of information, we utilize the tracking mechanism of YOLO to interpolate detections in frames that they are missing.
 - Class Interpolation: YOLO may classify the same object differently in different frames. For that cause we will take the most frequent classification that YOLO provides.
- 3) 2D to 3D projections - By setting a point in the scene as the origin, and gathering real distances along the scene, we calculate a homography matrix that can help us transform each pixel in the frame into a point in Euclidean space. The DAS sampling points, that assemble the fiber trajectory, are also coupled with 3D locations that are relative to the origin.
 - 4) Strain Rate Map Generation: By posing the fiber sampling points, along with the detected vehicles in 3D space, we can use formulas (1), (2) and (3) help us to calculate the approximated strain rate amplitude.
 - 5) Calibration Optimization - Using PyTorch platform for establishing computational graphs - it is possible to use variations of SGD to optimize the initial location estimated for the fiber.

IV. IMPLEMENTATION

A. DAS Signals Pre-Processing

The data utilized in our research was captured on March 21st, 2023. The recorded fiber was situated adjacent to a dual carriageway accommodating various modes of transportation, including cars, buses, motorcycles, and pedestrians. Furthermore, within the recorded road segment, there exists a designated bus stop. To the best of our knowledge, the fiber's alignment closely matches the direction of the road. These descriptions, together with additional factors

such as fiber medium coupling, have a notable impact on the data. Since our primary objective was to detect and classify vehicular movements, the pre-processing steps employed were specifically tailored to facilitate this task. As mentioned before, the data that was captured by our DAS system is shaped as a matrix with one axis corresponds with time and the other with distance. The two axes are affected by the spatial sampling rate, that was determined to be 1m^{-1} , and the temporal sampling rate, that was set to $1[\text{KHz}]$. This temporal sampling rate is not necessary for our research, as the relevant frequency band that carry vehicular movement signature are quasi-static. Thus, the data was down-sampled to $30[\text{Hz}]$, containing frequencies up to $10[\text{Hz}]$.

To eliminate signals that exhibit velocities inconsistent with vehicles and exclude waves with frequencies higher than 3Hz (2Hz for quasi-static waves with a safety margin), it is necessary to employ a two-dimensional filter. A one-dimensional filter alone cannot effectively address the underlying two-dimensional problem and fails to consider the signal speed along the x -axis. Thus, we utilized an F-K filter. F-K filtering involves the conversion of seismic data, typically represented in the time and displacement domain, into the frequency and wave number (F-K) domain. The seismic data is then subjected to filtering to eliminate undesired frequencies outside the desired seismic signal range. Subsequently, the data is converted back to the time-displacement domain. In simple terms, this filter consists of a two-dimensional Fourier transform, followed by a two-dimensional band-pass filter, and finally a reverse two-dimensional Fourier transform. Our use of the F-K filter was tailored to capture the characteristics of the road monitored in the data acquisition experiment. Hence, we eliminated any signals with a wave number (K) greater than 0.045, and any velocity outside the range of $[2.5, 25]$ m/s (corresponding to 9 – 90 km/hr). See figure 2 and figure 3 for a visualization of the F-K filter's impact on our DAS data, depicting 60 seconds of sampling over 90 channels.

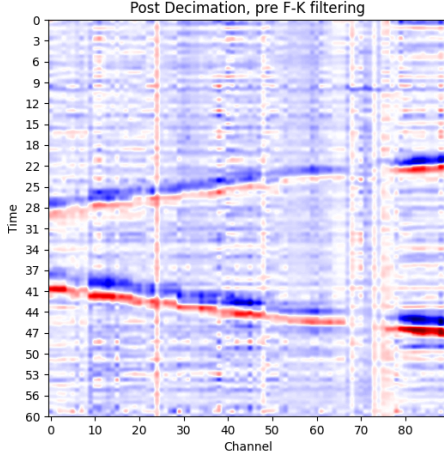


Fig. 2. DAS output post decimation and pre F-K filtering

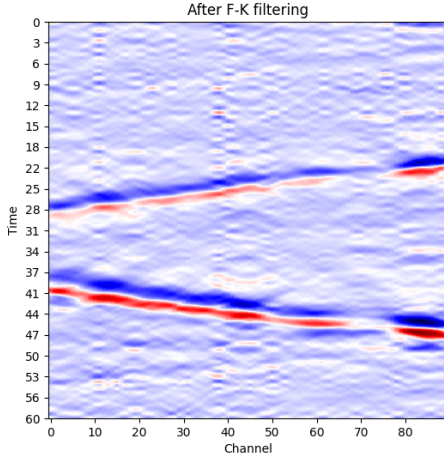


Fig. 3. DAS output post F-K filtering

B. Object Detection

The primary focus of our research revolves around the generation of synthetic strain rate maps that best resemble the DAS output data. Following the filter that was introduced in the previous subsection, we aim to detect, track, and classify the vehicles in the scene in each frame. For that purpose, we utilize YOLOv8, a state-of-the-art computer vision pre-trained model that can perform all of the above

tasks. When we apply the model, we document its results in a DB that includes the objects track ID, class and data regarding their location (in terms of pixels in the frame). Nevertheless, YOLOv8 has some drawbacks that can impair the strain rate map generations that we opt to overcome.

- 1) Handling Still Objects - YOLO detects all of the objects in a frame, but since the DAS interrogator measures strain rate - still objects do not affect it and should be excluded from further calculations. Hence, we've developed an algorithm that documents the data of moving objects only by keeping an associative array that couples every tracked object with its previous locations. Only if the displacement between the current detected location and the previous one goes over some preset threshold - the data of the object will be documented.
- 2) Missing detections - YOLOv8 often misses the detection of objects in some frames, resolving in discontinuity of the presence of vehicles in the processed data. This fault has fatal implications over the accuracy of the strain rate generation algorithm that we will later present, as continuity of the strain function is a necessary condition for its differentiability. To deal with this fault we utilize the YOLOv8 tracking mechanism. For each tracked object (that has distinct track ID) we look for its first and last frame of appearance, and linearly interpolate its location in missing frames.
- 3) Miss-classification - Despite its ability to track objects and even give them a distinct identifier, YOLOv8 may classify each object differently in each frame. Different classes (for example a private car versus a bus) have different mass that is later reflected in our calculations. To handle this fault we iterate over each of the tracked objects, identify the most frequent class that it was classified with, and then assign it to all of its instances in the detection DB.

C. Depth Estimation - Pixel to Euclidean Space Projections

Formulas (1), (2) and (3) depict the link between the spatial coordinates x, y, z and the strain rate. Hence the ability to pose both fiber and vehicles in relation to the same origin is crucial. Our first task in that manner would be to pose the vehicles along the scene. Since the experiment was conducted with monocular filming, we've utilized the homography matrix[2], to perform transformations from the pixel space to the Euclidean space.

The homography matrix is a 3×3 invertible matrix with 8 DoF and it is used to perform plane to plane transformations. In order to calculate it we had to provide real distance estimations of objects in the scene in relation to the origin. Moreover, it is worthy to note that the homography matrix construction is subjected to the assumption that the scene is approximately plane (i.e all vehicles travel at $z = 0$).

Once we have the homography matrix, we apply it over each vehicle detection to project its location in the frame to a 3D location in the real scene.

D. Strain Map Generation

A cardinal focus of our research was to generate the synthetic strain rate map out of a street scene video. Now that we have extracted all of the required information from the video we can formulate the following algorithm to calculate the strain rate map.

For that cause, we transform the information stored in the DB we have created into a tensor VT (video tensor) such that it couple each frame to the moving vehicles detected in it, and each vehicle is coupled to its approximated mass and location in the scene 3D space. In addition to that, we create a fiber-optic abstract object FO (fiber object) that couples each sampling channel with a 3D coordinate in the scene space.

Our algorithm relies on the following formulation, and uses the tensor and fiber-optic object above to calculate the strain rate map:

$$\begin{aligned} \dot{\epsilon}_{DAS} &= \lim_{h \rightarrow 0} \frac{\epsilon_{DAS}(x, y, z, t + h) - \epsilon_{DAS}(x, y, z, t - h)}{2h} \\ &\approx \frac{\epsilon_{DAS}(x, y, z, t + T) - \epsilon_{DAS}(x, y, z, t - T)}{2T} \end{aligned} \quad (4)$$

where $T = \frac{1}{fps}$

Algorithm 1 Strain Rate Calculation Using Fiber Optic Locations and Vehicle Detections

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1: Input:  $VT, FO$ 
2: Output:  $SR \in \mathbb{R}^{|frames| \times |channels|}$ 
3: Initialize a strain map  $S \in \mathbb{R}^{|frames| \times |channels|}$  and set its elements to 0.
4: Assume each vehicle has a 3D location and mass coupled to it
5: Begin:
6: for each frame in  $VT$  do
7:   for each vehicle in the frame do
8:     Transform vehicle 3D coordinates to fiber space coordinates (See Eq.1)
9:     for each channel in  $FO$  do
10:       $s \leftarrow calcStrain(vehicle, channel)$  (Eq.2)
11:       $S[frame, channel] \leftarrow S[frame, channel] + s$ 
12:     end for
13:   end for
14: end for
15:  $SR \leftarrow \frac{\partial S}{\partial t}$  (Eq.4)
16: Return  $SR$ 

```

V. OPTIMIZATION

In our system, we opt to optimize the coordinates x, y, z that are coupled to each channel. Our optimizer is a compound unit, with two phases of optimization: initialization stage, responsible for choosing the best initial fiber optic channel locations ; fine alignment optimization stage, responsible for optimizing channels locations and revealing the fiber's trajectory and curvature.

A. Fiber Optic Locations Initialization

Our empirical experience showed that not any guessed location of the fiber optic will converge to a reasonable location. The initialization has a great impact over the loss convergence and the integrity of the result. Hence, we developed a pre-optimization phase that is solely designated for choosing the best initial fiber optic channel

locations. In this phase we use normalized cross-correlation to be the metric we want to maximize. Equivalently, we opt to minimize the term:

$$\mathcal{L}_{\text{NCC}}(I, J) = 1 - \frac{\sum_i p_i \cdot g_i}{\sqrt{\sum_i p_i^2 \sum_i g_i^2}} \quad (5)$$

In this phase we iterate over possible points that define the borders of initialization of the fiber optic. Since the road in our filming goes across the x axis of the frame, we iterate over the y dimension - initializing each time the fiber optic locations and generate the strain map as if it was there. At the end of the iterative process we take the set of points that yielded the highest cross correlation with the DAS processed data.

B. Strain Rate Map Fine Alignment Optimization

The second stage of our optimizer focuses on the segmentation optimization of video-based generated strain maps, which are compared to DAS-based strain maps, treated as the ground truth. We use gradient based optimizer with annealing. Its forward step requires calculating the distances of each vehicle from the fiber and generate new video-based strain map. We define a set of regularizers and loss functions to guide the evaluation phase of the optimization process. In the following subsections, we provide a detailed explanation of these regularizers and loss functions. At section VI we provide the specifications regarding the optimizer configurations we use.

1) *Regularization*: To impose a physical solution on the optimizer, we have designed two regularizers that are added to the loss function and should be minimized.

a) *Integrity*: The sampling channels along in the fiber space are 1 meter distant away from each other. This implies that the euclidean distance between each two consecutive channels can be at most 1 meter. Hence if we denote $C \triangleq (\# \text{channels} - 1)$ and $\underline{d} \in \mathbb{R}^C$ to be a vector that holds these distances, we can formulate the regularizer:

$$R_{\text{integrity}} = \sum_{i \in C} \max(d_i - 1, 0) \quad (6)$$

To allow some flexibility, we set a lower bound and an upper bound for the regularizer. We denote the upper bound as 'UB' and the lower bound as 'LB'. Therefore, in practice, we use the following formula:

$$R_{\text{integrity}} \triangleq \sum_{i \in C} \max(d_i - UB, 0) + \max(LB - d_i, 0) \quad (7)$$

b) *Smoothness*: To get reasonable fiber trajectory in terms of curves, we define another regularizer denoted as $R_{\text{smoothness}}$. To calculate $R_{\text{smoothness}}$ we use Frenet-Serret formulas which can be used to describe the behavior of a space curve. One of Frenet-Serret formulas' parameters is κ which represents the curvature (measures how sharply the curve bends). κ depends on the tangent unit vector $\mathbf{T}$. If we denote the arc length as s we get that κ is given by:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| \quad (8)$$

$$\mathbf{T} = \frac{d\mathbf{r}}{ds} \quad (9)$$

For our purposes, we can simply set κ to be the Laplacian of the fiber's channels locations:

$$\kappa \triangleq \|\nabla^2 (\text{FOS}[\underline{x}, \underline{y}, \underline{z}])\|_1 \quad (10)$$

We also calculated the radius of curvature denoted as ρ and given by: $\rho = \frac{1}{\kappa}$. We want to limit ρ by preset threshold. This limitation implies component of $R_{\text{smoothness}}$ regularizer which we formulate as follows:

$$\rho_{\text{penalty}} \triangleq \max(0, \rho - \text{threshold}) \quad (11)$$

Overall, we can now formulate the regularizer:

$$R_{\text{smoothness}} \triangleq \kappa + \rho_{\text{penalty}} \quad (12)$$

Note: in both regularizers (integrity, smoothness) we use coefficients for each component to control the regularizers influence. To simplify the formulation we exclude them from the formulas.

2) *Loss*: In the strain map fine alignment optimization stage, our objective is to align the synthesized video-based strain maps with the structure of the ground truth strain maps derived from DAS data. On top of the normalized cross-correlation, we have found that Dice loss is particularly well-suited for this alignment task due to its focus on the overlap between predicted and ground truth regions of interest. We define a threshold that can discriminate regions with high absolute value of strain rate amplitude. We define the set of elements that surpassed the threshold in the predicted array as A and the corresponding set of elements in the ground truth as B . Then the Dice loss is formulated as follows:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2|A \cap B|}{|A| + |B|} \quad (13)$$

where A is the generated strain rate map and B is the ground truth strain rate map

However, we encountered some challenges during our empirical experience. Specifically, synthesized data can include outliers from incorrect detection of moving vehicles, while ground truth data may contain noise not present in the synthesized data. To deal with these challenges we use Huber loss which reduce the influence of outliers without giving them too significant attention, but also has a continuous and smooth gradient. The Huber loss is defined as:

$$\mathcal{L}_{\text{Huber}}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ \delta \cdot (|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \quad (14)$$

VI. RESULTS

We show the results of the proposed system for DAS and camera calibration. The scene takes place on a two way straight road. The road does not have significant turns, ups and downs. The camera has been set on a roof of a nearby building so there is a large distance and an angle between the camera and the road. The entire scene is about 20 minutes long, but we are going to focus on only 1 minute. Our use-case involves 2 buses driving in opposite directions (on different lanes) in different times (the

camera doesn't capture the moment they pass by each other). We emphasize this point because, based on Equations 1-3, driving in different lanes is essential to avoid obtaining a symmetric solution, as the single-lane problem lacks convexity. Section VI-F demonstrates that this one-minute segment provides sufficient data to accurately estimate the fiber locations.

A. DAS Signals Pre-Processing

After decimation and activating F-K filtering over the DAS data, we get the following strain map:

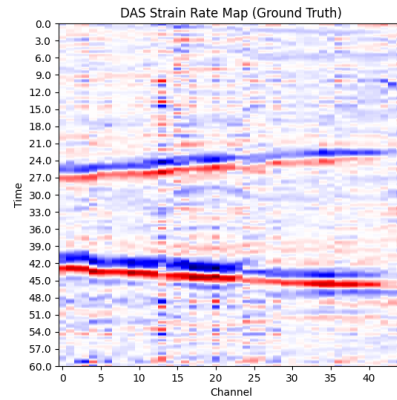


Fig. 4. DAS based strain map. Values are between [0,1]. Blue pixels represent lower values, as red pixels represent higher values.

As expected, we can see that at each channel, there are rise and fall of the strain rate values when the vehicle passing by the channel.

B. Fiber Optic Locations Initialization

We use YOLO to detect and track vehicles of the scene. Then we generate synthesized strain map based on the video. We start from guessed locations and run the initialization process (section V-A) to get better approximation of the initial fiber locations.

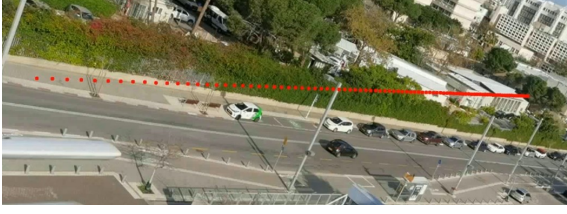


Fig. 5. Guessed fiber optic channels projected locations

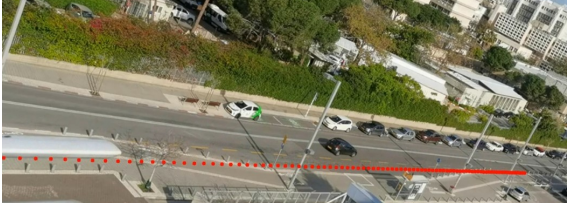


Fig. 6. Post initialization fiber optic channels projected locations

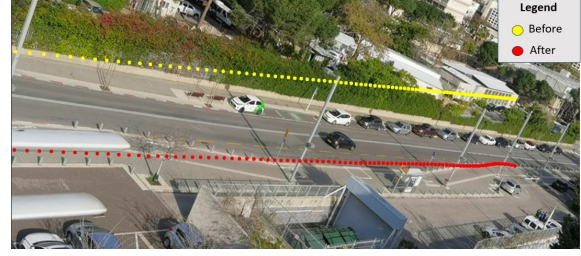


Fig. 7. Fiber optic channels projected locations before and after optimization process

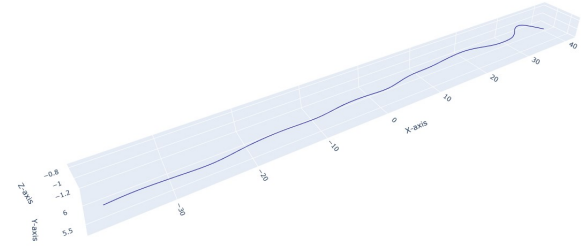


Fig. 8. Fiber optic channels 3D locations after optimization process

C. Strain Rate Map Fine Alignment Optimization

Now that we initialize the approximate fiber's channels locations, we can optimize the locations to get better estimation of the fiber structure and curvature. We run the second stage of our optimizer with the following parameters:

- Learning rate: 0.1
- Epochs: 2000
- Cosine annealing that reaches 0.001 with restarts every 200 epochs
- Integrity regularizer upper bound: 1.02 [m]
- Integrity regularizer lower bound: 0.75 [m]
- Regularizers coefficients: 5.0
- Loss functions coefficients: 1.0

This configuration allows significant changes in the segmentation map while maintaining the physical integrity of the solution. Below are visualizations of the results:

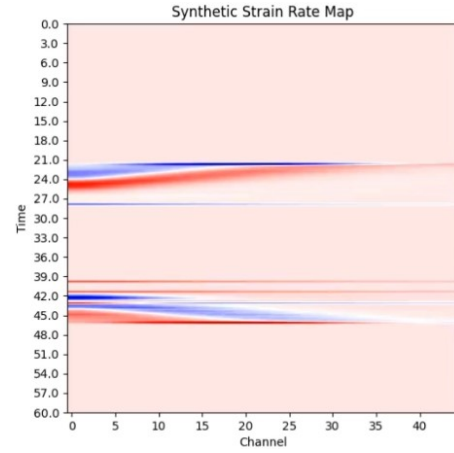


Fig. 9. Video based generated strain map with guessed fiber channels locations

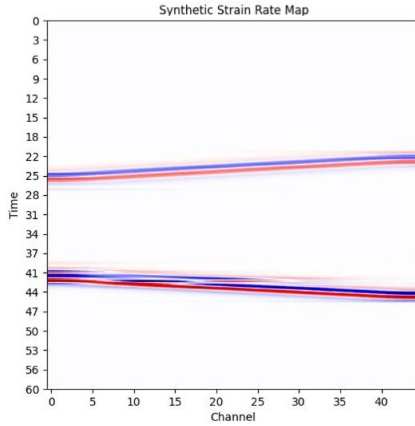


Fig. 10. Video based generated strain map with estimated fiber channels locations after optimization

Figures 7 and 8 present the final estimation of the fiber locations, showing that the optimizer successfully captured the fiber's curves. Figures 9 and 10 illustrate the significant changes in the generated strain map, which is now aligned with the DAS-based strain map at figure 4.

D. Evaluation

We evaluate the results using four methods: convergence of the segmentation alignment, channel locations comparison with manual calibration, loss comparison with manual calibration and a field tour with the infrastructure project manager from Tel Aviv University.

a) Convergence Graph: The convergence graph shows a rapid decrease in loss over the first 100 epochs. Due to the annealing strategy we employed, the convergence point shifts slightly until stabilizing. The loss, which includes the loss functions and regularizers presented at V-B2 and V-B1, decreases significantly, from 13.4 to 0.63, indicating a high degree of alignment between the two strain maps.

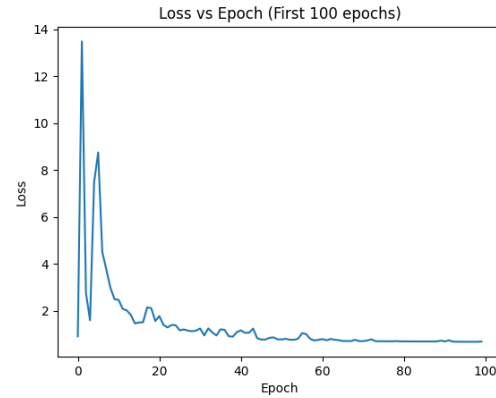


Fig. 11. Convergence graph for the first 100 epochs, showing a rapid decrease in loss.

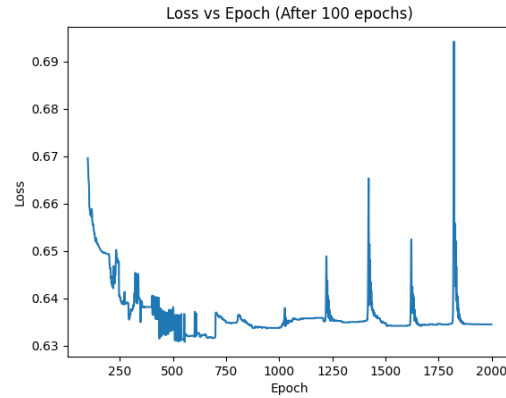


Fig. 12. Post-100 epoch convergence graph, showing stabilization of the convergence point despite the cosine with restarts annealing method

b) Fiber Locations Comparison with Manual Calibration Results: A comparison with manual calibration demonstrates that the automatic calibration not only closely aligns with the manual results—showing an average distance difference of approximately 1.8 meters—but also provides more granular details. The figure below illustrates the projected locations from both methods:

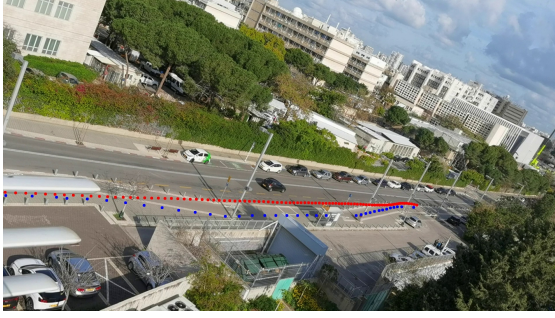


Fig. 13. Manual vs. automatic calibration. Manual calibration channel locations are in blue, and automatic calibration channel locations are in red.

Visual inspection reveals:

- The fiber locations from both methods are quite similar.
- The manual calibration is less informative, capturing only 36 out of the 90 channels that should have been measured.

Next, we calculated the distances between the fiber locations from the two methods. For each manually measured channel, we calculated the distance to the nearest auto-measured channel¹:

- Mean distance: 1.802 [m]
- Standard deviation: 0.911 [m]
- Maximum distance: 2.984 [m]
- Minimum distance: 0.437 [m]

c) Loss Comparison with Manual Calibration

Results: We apply the loss functions introduced in V-B2 to strain maps generated from both the manual and automated calibration results. Since manual calibration provides less detailed information, we use linear interpolation to produce higher-quality strain maps. Losses are compared across 9 different minutes,² where strain maps for both methods are generated and evaluated against the

¹An alternative approach could involve comparing the manually and automatically measured channel locations directly. However, this method would overlook the potential inaccuracies introduced by depth estimation, projection techniques, and possible errors in manual calibration.

²Despite using only a 1-minute scene, the automated calibration shows loss metrics similar to manual calibration across nine different intervals, suggesting a promising impact on performance. A detailed discussion of this topic is provided in Section VI-F

DAS-based ground-truth strain maps for the corresponding minute. The calculated losses for both methods were nearly identical. The results are shown below:

minute index	auto calibration loss	manual calibration loss
0	0.61	0.55
1	0.52	0.55
2	0.86	0.87
3	0.77	0.76
4	0.78	0.82
5	0.76	0.75
6	0.59	0.51
7	0.67	0.78
8	0.77	0.76

TABLE I

LOSS COMPARISON BETWEEN AUTOMATED AND MANUAL CALIBRATION METHODS OVER 9 MINUTES OF THE SCENE

Overall, the results from automatic calibration are consistent with those from the manual method. Both methods confirm that the fiber is located on the same sidewalk, and the estimated locations of channels closer to the camera are nearly identical. However, some variations in exact locations were observed, particularly near and beyond the bus station, which is farther from the camera. While the precise cause of these deviations is unclear, several potential factors could be contributing to the differences:

- The camera's position might have influenced the accuracy of location estimates for distant channels, affecting both calibration methods. For instance, in automated calibration, detection accuracy for distant vehicles was lower compared to those closer to the camera.
- Using a single camera impacts the quality of projection. As discussed in Section IV-C, a homography transformation is applied with the assumption of a constant z-axis in 3D coordinates. This assumption may cause slight deviations in the results.
- Manual calibration might not be as precise, and the channel numbering may be biased.

d) Field Tour With The Infrastructure Project Manager: During a field tour with the infrastructure project manager, we observed that the fiber is positioned closer to the road, near the sidewalk edge, and that there is a curve after the bus stop. These findings are consistent with our results.

E. Verify Physical Solution

We verify the solution is physically feasible in two manners: reasonable trajectory in terms of curves and distance between consecutive channels.

a) Distance Between Consecutive Channels:

The expected distance between consecutive channels is approximately 1 meter, with minimal deviation between consecutive channel pairs. Upon analyzing these distances we calculated:

- Mean distance: 0.87 [m]
- Standard deviation: 0.0076 [m]

The mean distance being close to 1 meter is consistent with our expectations. The slight deviation could be attributed to twists and curves between the channels that are not captured in the results. Additionally, the low standard deviation indicates minimal variation, as anticipated. Overall, the distances between consecutive channels are reasonable, supporting the physical validity of the solution.

b) *Fiber Trajectory and Curvature:* Verification of the fiber's trajectory and curvature reveals that the radius of curvature is significantly above the acceptable limit, with a calculated value of $\rho = 79130.92m$.

Assuming that the fiber's diameter is approximately 25-50 mm, the minimum allowable radius of curvature should range from 250-500 mm (according to the Fiber Optic Association Reference Guide). The resulting fiber trajectory exhibits a much larger radius of curvature, indicating that the solution is physically feasible in terms of bending constraints.

F. Performance And Analysis

In this section, we analyze the performance of our optimization algorithm by comparing results from a 1-minute scene involving two buses driving in opposite directions with those from a 9-minute scene with various vehicles, including buses and cars. To ensure a fair comparison, we filtered out all vehicles other than buses in the 9-minute scene. To compare the 9-minute scene with the 1-minute scene, we ran the second stage of the optimization process on the initialized fiber locations (see section V-A) with identical hyperparameters:

- Learning rate: 1

- Epochs: 10000
- Cosine annealing that reaches 0.001 and restarts every 1000 epochs
- Integrity regularizer upper bound: 1.02 [m]
- Integrity regularizer lower bound: 0.75 [m]
- Regularizers coefficients: 1.0
- Huber loss function coefficient: 1.0
- Dice loss function coefficient: 10.0

This configuration was designed to allow significant adjustments and emphasize the fiber structure and curves. We evaluated the results using the same comparison methods as in Section VI-D. Specifically, we first compared the fiber locations resulting from the two methods, and then we compared their respective losses.

a) Fiber Locations Comparison:



Fig. 14. Fiber channel locations for the two runs. Results from the 9-minute scene (with buses only) are shown in red, and results from the 1-minute scene are shown in yellow. As reference, manual calibration is depicted in blue.

Visual inspection reveals:

- The fiber locations of all methods are quite similar.
- Results from the 9-minute scene are more distant from manual calibration, with some channels located on the road.

We calculated the distances between the results based on 9 minutes scene to the manual calibration results. For each manually measured channel, we calculated the distance to the nearest auto-measured channel:

- Mean distances 2.438 [m]
- Standard deviation of the distances 1.013 [m]
- Maximum distance 4.041 [m]

- Minimum distance 0.437 [m]

Similarly, we calculated the distances between the results based on 1 minute scene to the manual calibration results:

- Mean distances 1.161 [m]
- Standard deviation of the distances 0.707 [m]
- Maximum distance 2.249 [m]
- Minimum distance 0.144 [m]

b) Loss Comparison: We applied the loss function presented at V-B2 over the generated strain maps based on the 9 minute calibration results and the 1 minute calibration results enables to compare the losses for all 9 minutes of both methods:

batch index	automated 9 minute	automated 1 minute
0	0.61	0.92
1	0.60	0.93
2	0.88	0.86
3	0.72	0.67
4	0.76	0.77
5	0.77	0.78
6	0.65	0.59
7	0.80	0.69
8	0.81	0.76

TABLE II
COMPARISON OF AUTOMATED 9 MINUTE CALIBRATION AND
AUTOMATED 1 MINUTE CALIBRATION LOSS FOR EACH
MINUTE

Observations:

- Between minutes 2 and 5, the losses for both methods are similar.
- Between minutes 6 and 8, the losses from the 1-minute scene are lower than those from the 9-minute scene.
- At minutes 0 and 1, the losses from the 9-minute scene are lower than those from the 1-minute scene.

Overall, the losses are comparable, showing no clear advantage for either method in terms of loss calculations. These results suggest that high-quality, smaller datasets can yield results comparable to, or even better than, those from larger datasets. This conclusion has significant implications for performance, user experience, and the requirements of our approach, emphasizing that smaller, high-quality datasets can be as effective, if not more effective, than larger ones.

VII. DISCUSSION AND CONCLUSIONS

This paper focuses on the spatial calibration between DAS and camera systems. Our proposed solution builds upon previous research in traffic analysis using DAS. In this work, we developed a system capable of analyzing a short video recording of street traffic alongside corresponding DAS measurements from a nearby fiber. The system can accurately reconstruct the fiber's trajectory, achieving a deviation of only 1-2 meters over distant channels relative to the camera, and essentially providing a proof of concept for this purpose. Additionally, our system can determine the fiber's trajectory within minutes, using only a few minutes of informative scene data (video and DAS). In fact, we demonstrated that just 1 minute of well-captured data is sufficient to achieve quite accurate results. We focused on data involving bus traffic due to the high SNR produced by seismic waves from buses, which generate noticeable strains on the fiber's strain map. Future work could examine the limitations of our system, particularly whether data from car traffic alone can also achieve accurate results. Further improvements could be made by refining the recording settings, such as optimizing the camera angle or using stereo recordings to improve depth estimation. Additionally, our system does not account for the coiling of optical fibers, which is a common practice during deployment. Addressing this challenge could enhance the system's ability to generalize and ensure the full integrity of the calibration process. Furthermore, our system can be used to augment data for other DAS applications by mass-producing synthesized DAS data that can be consistent with a specific (real, or fake) fiber-optic deployment. We believe that our novel solution to the spatial calibration problem will serve as a foundation for any future use of fiber optic sensing technologies.

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